Bibliography

[Abd17] A. R. Abdulghany. "Generalization of parallel axis theorem for rotational inertia". In: *American Journal of Physics* 85 (Oct. 2017), pp. 791–795.
DOI: 10.1119/1.4994835.

[ACFL95] E. F. Arias, P. Charlot, M. Feissel, and J.-F. Lestrade. "The extragalactic reference system of the International Earth Rotation Service, ICRS". In: *Astronomy & Astrophysics* 303 (Nov. 1995), pp. 604–608.

[ACM06] I. Alonso-Mallo, B. Cano, and M. J. Moreta. "Stability of Runge–Kutta–Nyström methods". In: *Journal of Computational and Applied Mathematics* 189.1-2 (1st May 2006), pp. 120–131.
DOI: 10.1016/j.cam.2005.01.005.

[AK15] Z. A. Anastassi and A. A. Kosti. "A 6(4) optimized embedded Runge–Kutta–Nyström pair for the numerical solution of periodic problems". In: *Journal of Computational and Applied Mathematics* 275 (2015), pp. 311–320.
DOI: 10.1016/j.cam.2014.07.016.

[AM69] M. F. Atiyah and I. G. Macdonald. *Introduction to Commutative Algebra*. Addison-Wesley, 1969.
eprint: https://webhomes.maths.ed.ac.uk/~v1ranick/papers/atiyahmacdonald.pdf.

[And17] "Andrew". "Davenport's Q-method (Finding an orientation matching a set of point samples)". Mathematics Stack Exchange at https://math.stackexchange.com/q/2275087. 10th May 2017.

[Arc+11] B. A. Archinal, M. F. A'Hearn, E. Bowell, A. Conrad, G. J. Consolmagno, R. Courtin, T. Fukushima, D. Hestroffer, J. L. Hilton, G. A. Krasinsky, G. Neumann, J. Oberst, P. K. Seidelmann, P. Stooke, D. J. Tholen, P. C. Thomas, and I. P. Williams. "Report of the IAU Working Group on Cartographic Coordinates and Rotational Elements: 2009". In: *Celestial Mechanics and Dynamical Astronomy* 109.2 (2011), pp. 101–135.
DOI: 10.1007/s10569-010-9320-4.
eprint: http://goo.gl/d6WdlJ.

[Bab27] C. Babbage. *Table of Logarithms of the Natural Numbers, from 1 to 108000*. J. Mawman, 1827.
eprint: https://books.google.co.uk/books?id=64o_AAAAcAAJ.

[Bat38] H. Bateman. "Halley's methods for solving equations". In: *The American Mathematical Monthly* 45.1 (Jan. 1938), pp. 11–17.
DOI: 10.2307/2303467.

[BC72] R. A. Broucke and P. J. Cefola. "On the Equinoctial Orbit Elements". In: *Celestial Mechanics* 5.3 (May 1972), pp. 303–310.
DOI: 10.1007/BF01228432.

[BCM07] S. Blanes, F. Casas, and A. Murua. "Splitting methods for nonautonomous linear systems". In: *International Journal of Computer Mathematics* 84.6 (2007), pp. 713–727.
DOI: 10.1080/00207160701458567.

[BCR00] S. Blanes, F. Casas, and J. Ros. "Processing Symplectic Methods for Near-Integrable Hamiltonian Systems". In: *Celestial Mechanics and Dynamical Astronomy* 77.1 (2000), pp. 17–36.
DOI: 10.1023/A:1008311025472.

[BCR01a] S. Blanes, F. Casas, and J. Ros. "High-order Runge–Kutta–Nyström geometric methods with processing". In: *Applied Numerical Mathematics* 39.3–4 (2001). Themes in Geometric Integration, pp. 245–259.
DOI: 10.1016/S0168-9274(00)00035-0.

[BCR01b] S. Blanes, F. Casas, and J. Ros. “New Families of Symplectic Runge–Kutta–Nyström Integration Methods”. In: *Numerical Analysis and Its Applications*. NAA 2000 (Rousse, Bulgaria, 11th–15th June 2000). Ed. by L. Vulkov, P. Yalamov, and J. Waśniewski. Lecture Notes in Computer Science 1988. International Conference on Numerical Analysis and Its Applications. Springer Berlin Heidelberg, 2001, pp. 102–109.
DOI: 10.1007/3-540-45262-1_13.

[BCR99] S. Blanes, F. Casas, and J. Ros. “Symplectic Integration with Processing: A General Study”. In: *SIAM Journal on Scientific Computing* 21.2 (1999), pp. 711–727.
DOI: 10.1137/S1064827598332497.

[BDL09] S. Boldo, M. Daumas, and R.-C. Li. “Formally Verified Argument Reduction with a Fused Multiply-Add”. In: *IEEE Transactions on Computers* 58.8 (Aug. 2009), pp. 1139–1145.
DOI: 10.1109/TC.2008.216.

[Běi13] 北斗卫星导航系统公开服务性能规范（1.0 版）—*BeiDou Navigation Satellite System Open Service Performance Standard (Version 1.0)*. BDS-OS-PS-1.0. Chinese: http://www.beidou.gov.cn/zt/zcfg/201710/P020171202709828404659.pdf—English: http://www.beidou.gov.cn/xt/gfxz/201805/P020180509584193805984.pdf. 中国卫星导航系统管理办公室—China Satellite Navigation Office, Dec. 2013.

[Ben65] A. Bennett. “Characteristic Exponents of the Five Equilibrium Solutions in the Elliptically Restricted Problem”. In: *Icarus* 4.2 (May 1965), pp. 177–187.
DOI: 10.1016/0019-1035(65)90060-6.

[Ben97] J. R. Benada. *PO.DAAC Merged GDR (TOPEX-Poseidon) Generation B User’s handbook, version 2.0*. D-11007. Jet Propulsion Laboratory, 30th July 1997.
eprint: https://podaac-tools.jpl.nasa.gov/drive/files/allData/topex/L2/mgdrb/docs/uhmgdrb/html/usr_toc.htm.

[Bes52] F. W. Bessel. “Bessel an Olbers. Königsberg, 23. April 1818”. In: *Briefwechsel zwischen W. Olbers und F. W. Bessel*. Ed. by A. Erman. Vol. 2. 260. 1852, pp. 84–90.

[Beu03] H. Beust. “Symplectic integration of hierarchical stellar systems”. In: *Astronomy & Astrophysics* 400 (Mar. 2003), pp. 1129–1144.
DOI: 10.1051/0004-6361:20030065.

[BF54] P. F. Byrd and M. D. Friedman. *Handbook of elliptic integrals for engineers and physicists*. Springer-Verlag, 1954.

[BGM17] S. Boldo, S. Graillat, and J.-M. Muller. “On the Robustness of the 2Sum and Fast2Sum Algorithms”. In: *ACM Transactions on Mathematical Software* 44.1 (July 2017), pp. 1–14.
DOI: 10.1145/3054947.

[BJ10] X. Bai and J. L. Junkins. “Solving initial value problems by the Picard-Chebyshev method with NVIDIA GPUs”. In: *Advances in the Astronautical Sciences* 136.2 (2010), pp. 1459–1476.

[BJ12] X. Bai and J. L. Junkins. “Modified Chebyshev-Picard iteration methods for solution of initial value problems”. In: *The Journal of the Astronautical Sciences* 59.1 (2012), pp. 327–351.
DOI: 10.1007/s40295-013-0021-6.

[Bjö94] Å. Björck. “Numerics of Gram-Schmidt orthogonalization”. In: *Linear Algebra and its Applications* 197–198 (1994), pp. 297–316.
DOI: 10.1016/0024-3795(94)90493-6.

[BL69] P. F. Byrd and K. L. Lee. *Chebyshev series solution of nonlinear ordinary differential equations-Initial-value problems*. Tech. rep. 19690027545. 1969.

[Bla+13] S. Blanes, F. Casas, A. Farrés, J. Laskar, J. Makazaga, and A. Murua. “New families of symplectic splitting methods for numerical integration in dynamical astronomy”. In: *Applied Numerical Mathematics* 68 (2013), pp. 58–72.
DOI: 10.1016/j.apnum.2013.01.003.

[BM02] S. Blanes and P. C. Moan. “Practical Symplectic Partitioned Runge–Kutta and Runge–Kutta–Nyström Methods”. In: *Journal of Computational and Applied Mathematics* 142.2 (May 2002), pp. 313–330.
DOI: 10.1016/S0377-0427(01)00492-7.

[Boy06] J. P. Boyd. "Computing real roots of a polynomial in Chebyshev series form through subdivision". In: *Applied Numerical Mathematics* 56.8 (2006), pp. 1077–1091. DOI: 10.1016/j.apnum.2005.09.007.

[Boy13] J. P. Boyd. "Finding the Zeros of a Univariate Equation: Proxy Rootfinders, Chebyshev Interpolation, and the Companion Matrix". In: *SIAM Review* 55.2 (2013), pp. 375–396. DOI: 10.1137/110838297.

[Bra+14] B. K. Bradley, B. A. Jones, G. Beylkin, K. Sandberg, and P. Axelrad. "Bandlimited implicit Runge–Kutta integration for Astrodynamics". In: *Celestial Mechanics and Dynamical Astronomy* 119.2 (2014), pp. 143–168. DOI: 10.1007/s10569-014-9551-x.

[Bra06] C. J. F. ter Braak. "A Markov Chain Monte Carlo version of the genetic algorithm Differential Evolution: easy Bayesian computing for real parameter spaces". In: *Statistics and Computing* 16 (Sept. 2006), pp. 239–249. DOI: 10.1007/s11222-006-8769-1.

[Bre73] R. P. Brent. *Algorithms for minimization without derivatives*. Prentice-Hall, 1973.

[Bru70] C. Bruhns. *Neues logarithmisch-trigonometrisches Handbuch auf sieben Decimalen—A New Manual of Logarithms to Seven Places of Decimals—Nouveau manuel de logarithmes à sept décimales pour les nombres et les fonctions trigonométriques—Nuovo manuale logaritmico-trigonometrico con sette decimali*. German: https://books.google.de/books?id=GglTAAAAcAAJ—English: https://books.google.de/books?id=v-cGAAAAYAAJ—French: https://books.google.de/books?id=RXhNAAAAYAAJ. Bernhard Tauchnitz, 1870. German, French, English, and Italian versions published simultaneously; we have not been able to find a facsimile of the Italian edition.

[BS96] F. Blanc and C. Schgounn. *AVISO/Altimetry, AVISO User Handbook for Merged TOPEX/-POSEIDON products, Edition 3.0*. AVI-NT-02-101-CN. Centre national d'études spatiales, July 1996.

[BSFL98] R. S. Bhat, B. E. Shapiro, R. B. Frauenholz, and R. K. Leavitt. *TOPEX/Poseidon Orbit Maintenance for First Five Years*. Technical Report 97-1398. Jet Propulsion Laboratory, 11th May 1998. eprint: https://trs.jpl.nasa.gov/handle/2014/22860.

[Bul65] R. Bulirsch. "Numerical Calculation of Elliptic Integrals and Elliptic Functions". In: *Numerische Mathematik* 7.1 (1st Feb. 1965), pp. 78–90. DOI: 10.1007/BF01397975.

[Bul69] R. Bulirsch. "Numerical Calculation of Elliptic Integrals and Elliptic Functions. III". In: *Numerische Mathematik* 13.4 (Aug. 1969), pp. 305–315. DOI: 10.1007/BF02165405.

[Can18a] S. Canon. "A trick I've used for years and should write up". Tweets at https://twitter.com/stephentyrone/status/1016283784067665920 *sqq.*, with a note at https://twitter.com/stephentyrone/status/1016328842296864770. 9th July 2018.

[Can18b] S. Canon. "A trick I've used for years and should write up: Quick version". Tweets at https://twitter.com/stephentyrone/status/1057788315699687424 *sqq.* 1st Nov. 2018.

[Can61] M. Cantor. "Note historique sur l'extraction abrégée de la racine carrée". In: *Nouvelles annales de mathématiques. Journal des candidats aux écoles polytechnique et normale* 1.20 (1861). Ed. by O. Terquem and C.-C. Gerono, pp. 46–47. eprint: http://www.numdam.org/item/?id=NAM_1861_1_20__46_1.

[Cap12] M. Capderou. *Satellites : de Kepler au GPS*. Springer-Verlag France, 2012.

[CFSZ07] E. Celledoni, F. Fassò, N. Säfström, and A. Zanna. "The exact computation of the free rigid body motion and its use in splitting methods". Preprint at https://www.math.unipd.it/~fasso/research/papers/norway.pdf. Oct. 2007.

[CFSZ08] E. Celledoni, F. Fassò, N. Säfström, and A. Zanna. "The exact computation of the free rigid body motion and its use in splitting methods". In: *SIAM Journal on Scientific Computing* 30 (May 2008), pp. 2084–2112.

[CGM08] M. P. Calvo, S. González-Pinto, and J. I. Montijano. "Global error estimation based on the tolerance proportionality for some adaptive Runge–Kutta codes". In: *Journal of Computational and Applied Mathematics* 218.2 (2008). The Proceedings of the Twelfth International Congress on Computational and Applied Mathematics, pp. 329–341. DOI: 10.1016/j.cam.2007.02.034.

[Cha73] J. F. Chandler. "Determination of the Dynamical Properties of the Jovian System by Numerical Analysis". PhD thesis. Massachusetts Institute of Technology, 1973. eprint: http://hdl.handle.net/1721.1/51410.

[Cha95] J. Chapront. "Representation of planetary ephemerides by frequency analysis. Applications to the five outer planets". In: *Astronomy & Astrophysics – Supplement Series* 109 (Jan. 1995), pp. 181–192.

[Chi97] S. A. Chin. "Symplectic integrators from composite operator factorizations". In: *Physics Letters A* 226.6 (1997), pp. 344–348. DOI: 10.1016/S0375-9601(97)00003-0.

[CHMR96] M. P. Calvo, D. J. Higham, J. I. Montijano, and L. Rández. "Global error estimation with adaptive explicit Runge–Kutta methods". In: *IMA Journal of Numerical Analysis* 16.1 (1996), pp. 47–63. DOI: 10.1093/imanum/16.1.47.

[CHO73] C. J. Cohen, E. C. Hubbard, and C. Oesterwinter. *Astronomical Papers Prepared for the Use of the American Ephemeris and Nautical Almanac – Elements of the Outer Planets for One Million Years*. Vol. XXII. I. United States Government Printing Office, 1973.

[CK00] S. A. Chin and D. W. Kidwell. "Higher-order force gradient symplectic algorithms". In: *Physical Review E* 62 (Dec. 2000), p. 8746. DOI: 10.1103/PhysRevE.62.8746. eprint: http://arxiv.org/abs/physics/0006082.

[CN63] C. W. Clenshaw and H. J. Norton. "The solution of nonlinear ordinary differential equations in Chebyshev series". In: *The Computer Journal* 6.1 (1963), pp. 88–92. DOI: 10.1093/comjnl/6.1.88.

[Coh51] A. C. Cohen Jr. "Estimating Parameters of Logarithmic-Normal Distributions by Maximum Likelihood". In: *Journal of the American Statistical Association* 46.254 (June 1951), pp. 206–212. DOI: 10.2307/2279920.

[Col92] P. Colwell. "Bessel Functions and Kepler's Equation". In: *The American Mathematical Monthly* 99.1 (Jan. 1992), pp. 45–48. DOI: 10.2307/2324547.

[CR91] J. Candy and W. Rozmus. "A symplectic integration algorithm for separable Hamiltonian functions". In: *Journal of Computational Physics* 92.1 (Jan. 1991), pp. 230–256. DOI: 10.1016/0021-9991(91)90299-Z.

[CS93a] M. P. Calvo and J. M. Sanz-Serna. "High-Order Symplectic Runge–Kutta–Nyström Methods". In: *SIAM Journal on Scientific Computing* 14.5 (1993), pp. 1237–1252. DOI: 10.1137/0914073.

[CS93b] M. P. Calvo and J. M. Sanz-Serna. "The Development of Variable-step Symplectic Integrators with Application to the Two-body Problem". In: *SIAM Journal on Scientific Computing* 14.4 (July 1993), pp. 936–952. DOI: 10.1137/0914057.

[CV84] J. Chapront and D. T. Vu. "A new compact representation of ephemerides: application to the motion of Pluto, the Sun and the Galilean satellites of Jupiter". In: *Astronomy & Astrophysics* 141 (1984), pp. 131–143.

[Dek71] T. J. Dekker. "A Floating-Point Technique for Extending the Available Precision". In: *Numerische Mathematik* 18 (1971), pp. 224–242. DOI: 10.1007/BF01397083.

[Del90] J.-B. Delambre. "De l'usage du calcul différentiel dans la construction des tables astronomiques". In: *Mémoires de l'Académie Royale des Sciences de Turin* V (1790), pp. 143–180. eprint: https://www.biodiversitylibrary.org/item/32318#page/699/mode/1up.

[DEP87a] J. R. Dormand, M. E. A. El-Mikkawy, and P. J. Prince. "Families of Runge–Kutta–Nyström Formulae". In: *IMA Journal of Numerical Analysis* 7.2 (1987), pp. 235–250.
DOI: 10.1093/imanum/7.2.235.

[DEP87b] J. R. Dormand, M. E. A. El-Mikkawy, and P. J. Prince. "High-Order Embedded Runge–Kutta–Nyström Formulae". In: *IMA Journal of Numerical Analysis* 7.4 (1987), pp. 423–430.
DOI: 10.1093/imanum/7.4.423.

[DL42] G. C. Danielson and C. Lánczos. "Some improvements in practical Fourier analysis and their application to x-ray scattering from liquids". In: *Journal of the Franklin Institute* 233.4 (1942), pp. 365–380.
DOI: 10.1016/S0016-0032(42)90767-1.

[DM11] F. Diele and C. Marangi. "Explicit symplectic partitioned Runge–Kutta–Nyström methods for non-autonomous dynamics". In: *Applied Numerical Mathematics* 61.7 (2011), pp. 832–843.
DOI: 10.1016/j.apnum.2011.02.003.

[Dor96] J. R. Dormand. *Numerical Methods for Differential Equations – A Computational Approach*. CRC Press, 1996.

[DP78] J. R. Dormand and P. J. Prince. "New Runge-Kutta Algorithms for Numerical Simulation in Dynamical Astronomy". In: *Celestial mechanics* 18.3 (1978), pp. 223–232.
DOI: 10.1007/BF01230162.

[DP80] J. R. Dormand and P. J. Prince. "A family of embedded Runge-Kutta formulae". In: *Journal of Computational and Applied Mathematics* 6.1 (1980), pp. 19–26.
DOI: 10.1016/0771-050X(80)90013-3.

[DP86] J. R. Dormand and P. J. Prince. "A reconsideration of some embedded Runge-Kutta formulae". In: *Journal of Computational and Applied Mathematics* 15.2 (1986), pp. 203–211.
DOI: 10.1016/0377-0427(86)90027-0.

[DRT01] M. Daumas, L. Rideau, and L. Théry. "A Generic Library for Floating-Point Numbers and Its Application to Exact Computing". In: *Theorem Proving in Higher Order Logics*. 14th International Conference on Theorem Proving in Higher Order Logics (Edinburgh, Scotland, UK, 3rd–6th Sept. 2001). Ed. by R. J. Boulton and P. B. Jackson. 2152. Springer Berlin Heidelberg, 2001, pp. 169–184.
DOI: 10.1007/3-540-44755-5_13.

[DS78] L. C. W. Dixon and G. P. Szegő. *Towards Global Optimization 2*. North-Holland Publishing Company, 1978.

[Eme07] V. V. Emel'yanenko. "A method of symplectic integrations with adaptive time-steps for individual Hamiltonians in the planetary N-body problem". In: *Celestial Mechanics and Dynamical Astronomy* 98.3 (2007), pp. 191–202.
DOI: 10.1007/s10569-007-9077-6.

[ER03] M. El-Mikkawy and E. D. Rahmo. "A new optimized non-FSAL embedded Runge–Kutta–Nyström algorithm of orders 6 and 4 in six stages". In: *Applied Mathematics and Computation* 145.1 (2003), pp. 33–43.
DOI: 10.1016/S0096-3003(02)00436-8.

[Est60] G. Estrin. "Organization of computer systems: the fixed plus variable structure computer". In: *Papers Presented at the May 3-5, 1960, Western Joint IRE-AIEE-ACM Computer Conference*. Western Joint IRE-AIEE-ACM Computer Conference (San Francisco, California, 3rd–5th May 1960). Association for Computing Machinery, 1960, pp. 33–40.
DOI: 10.1145/1460361.1460365.

[Eve85] E. Everhart. "An Efficient Integrator that Uses Gauss-Radau Spacings". In: *Dynamics of Comets: Their Origin and Evolution*. 83rd Colloquium of the International Astronomical Union (Rome, Italy, 11th–15th June 1984). Ed. by A. Carusi and G. B. Valsecchi. Astrophysics and Space Science Library 115. International Astronomical Union. Springer Netherlands, 1985, pp. 185–202.
DOI: 10.1007/978-94-009-5400-7_17.

[Fan33] T. Fantet de Lagny. “Analyse Générale, ou Méthodes nouvelles pour résoudre les Problêmes de tous les Genres & de tous les Degrez à l’infini”. In: *Recueil des Mémoires de l’Académie Royale des Sciences depuis 1666 jusqu’à 1699*. Ed. by C. Richer. Vol. XI. Par la compagnie des libraires, 1733.
eprint: https://books.google.fr/books?id=KwP-kH6gmlEC.

[Fan91a] T. Fantet de Lagny. “Nouvelle methode de Mr. T. F. de Lagny pour l’approximation des Racines cubiques”. In: *Le Journal des sçavans* 1691.17 (14th May 1691), pp. 200–203.
eprint: https://gallica.bnf.fr/ark:/12148/bpt6k56538h/f202.double.

[Fan91b] T. Fantet de Lagny. *Méthode nouvelle, infiniment générale et infiniment abrégée, Pour l’Extraction des Racines quarrées, cubiques, &c. & pour l’Approximation des mêmes Racines à l’infini dans toutes sortes d’égalitez. Proposée à examiner aux Mathématiciens de l’Europe*. De l’Imprimerie d’Antoine Lambin, ruë S. Jacques, au Miroir, 1691.
eprint: https://gallica.bnf.fr/ark:/12148/bpt6k1039787.

[Fan92] T. Fantet de Lagny. *Methodes nouvelles et abbregées pour l’extraction et l’approximation des racines. Et pour resoudre par le cercle et la ligne droite, plusieurs problêmes solides & sursolides ; comme la duplication du cube, l’invention de deux & de quatre moyennes proportionnelles, &c. dans toute la précision possible, & d’une maniere praticable. Avec une dissertation sur les methodes d’arithmetique & d’analyse ; où l’on établit des principes generaux pour en juger*. De l’Imprimerie de Jean Cusson, ruë saint Jacques, à l’Image de saint Jean Baptiste, 1692.
eprint: https://nubis.univ-paris1.fr/ark:/15733/3415.

[Fan97] T. Fantet de Lagny. *Nouveaux elemens d’arithmetique et d’algebre, ou introduction aux mathematiques*. Chez Jean Jombert, prés des Augustins, à l’Image Nôtre-Dame, 1697.
eprint: https://books.google.fr/books?id=IbTtzq_fixAC.

[Far+13] A. Farrés, J. Laskar, S. Blanes, F. Casas, J. Makazaga, and A. Murua. “High precision symplectic integrators for the Solar System”. In: *Celestial Mechanics and Dynamical Astronomy* 116.2 (2013), pp. 141–174.
DOI: 10.1007/s10569-013-9479-6.

[Fin87] J. M. Fine. “Low order practical Runge–Kutta–Nyström methods”. In: *Computing* 38.4 (1987), pp. 281–297.
DOI: 10.1007/BF02278707.

[FMLG08] A. Fienga, H. Manche, J. Laskar, and M. Gastineau. “INPOP06: a new numerical planetary ephemeris”. In: *Astronomy & Astrophysics* 477.1 (2008), pp. 315–327.
DOI: 10.1051/0004-6361:20066607.

[FN83] T. Feagin and P. Nacozy. “Matrix formulation of the Picard method for parallel computation”. In: *Celestial mechanics* 29.2 (1983), pp. 107–115.
DOI: 10.1007/BF01232802.

[Fol+14] W. M. Folkner, J. G. Williams, D. H. Boggs, R. S. Park, and P. Kuchynka. “The Planetary and Lunar Ephemerides DE430 and DE431”. In: *Interplanetary Network Progress Report* 42.196 (2014).
eprint: http://ilrs.gsfc.nasa.gov/docs/2014/196C.pdf.

[Fon34] B. de Fontenelle. “Éloge de M. de Lagny”. In: *Histoire de l’Académie royale des sciences avec les mémoires de mathématique et de physique tirés des registres de cette Académie* 1734 (1734), pp. 107–114.
eprint: https://gallica.bnf.fr/ark:/12148/bpt6k3531x/f115.

[Fon58] B. L. B. de Fontenelle. *Œuvres de Monsieur de Fontenelle, Des Académies, Françoise, des Sciences, des Belles-Lettres, de Londres, de Nancy, de Berlin & de Rome*. Nouvelle édition. 11 vols. B. Brunet, 1758–1766.
DOI: 10.3931/e-rara-25886.

[For88] B. Fornberg. “Generation of Finite Difference Formulas on Arbitrarily Spaced Grids”. In: *Mathematics of Computation* 51.184 (Oct. 1988), pp. 699–706.
DOI: 10.1090/S0025-5718-1988-0935077-0.

[FR90] E. Forest and R. D. Ruth. “Fourth-order symplectic integration”. In: *Physica D: Nonlinear Phenomena* 43.1 (May 1990), pp. 105–117.
DOI: 10.1016/0167-2789(90)90019-L.

[Fuk09a] T. Fukushima. "Fast computation of complete elliptic integrals and Jacobian elliptic functions". In: *Celestial Mechanics and Dynamical Astronomy* 105.4 (25th Oct. 2009), pp. 305–328.
DOI: 10.1007/s10569-009-9228-z.

[Fuk09b] T. Fukushima. "Fast computation of Jacobian elliptic functions and incomplete elliptic integrals for constant values of elliptic parameter and elliptic characteristic". In: *Celestial Mechanics and Dynamical Astronomy* 105.1 (Oct. 2009), pp. 245–260.
DOI: 10.1007/s10569-008-9177-y.

[Fuk11a] T. Fukushima. "Precise and fast computation of the general complete elliptic integral of the second kind". In: *Mathematics of Computation* 80 (Feb. 2011), pp. 1725–1743.
DOI: 10.1090/S0025-5718-2011-02455-5.

[Fuk11b] T. Fukushima. "Precise and fast computation of a general incomplete elliptic integral of second kind by half and double argument transformations". In: *Journal of Computational and Applied Mathematics* 235.14 (May 2011), pp. 4140–4148.
DOI: 10.1016/j.cam.2011.03.004.

[Fuk12a] T. Fukushima. *xgscd.txt (Fortran program package to compute the Jacobian elliptic functions, sn(u|m), cn(u|m), dn(u|m))*. Dec. 2012.

[Fuk12b] T. Fukushima. "Precise and fast computation of a general incomplete elliptic integral of third kind by half and double argument transformations". In: *Journal of Computational and Applied Mathematics* 236.7 (Jan. 2012), pp. 1961–1975.
DOI: 10.1016/j.cam.2011.11.007.

[Fuk18] T. Fukushima. *xelbdj.txt: Fortran test driver for "elbdj"/"relbdj", subroutines to compute the double/single precision general incomplete elliptic integrals of all three kinds*. Software. Jan. 2018.
DOI: 10.13140/RG.2.2.11113.80489.

[Fuk97] T. Fukushima. "Picard iteration method, Chebyshev polynomial approximation, and global numerical integration of dynamical motions". In: *Astronomical Journal* 113 (1997), pp. 1909–1914.
DOI: 10.1086/118404.

[Gal86] S. Gal. "Computing elementary functions: A new approach for achieving high accuracy and good performance". In: *Accurate Scientific Computations* (Bad Neuenahr, Federal Republic of Germany, 12th–14th Mar. 1985). Ed. by W. L. Miranker and R. A. Toupin. Lecture Notes in Computer Science 235. Springer Berlin Heidelberg, Sept. 1986, pp. 1–16.
DOI: 10.1007/3-540-16798-6_1.

[Gan85] W. Gander. "On Halley's Iteration Method". In: *The American Mathematical Monthly* 92.2 (Feb. 1985), pp. 131–134.
DOI: 10.2307/2322644.

[GB91] S. Gal and B. Bachelis. "An Accurate Elementary Mathematical Library for the IEEE Floating Point Standard". In: *ACM Transactions on Mathematical Software* 17.1 (Mar. 1991), pp. 26–45.

[Gen72a] W. M. Gentleman. "Algorithm 424: Clenshaw-Curtis quadrature [D1]". In: *Communications of the ACM* 15.5 (May 1972), pp. 353–355.
DOI: 10.1145/355602.355603.

[Gen72b] W. M. Gentleman. "Implementing Clenshaw-Curtis quadrature, I methodology and experience". In: *Communications of the ACM* 15.5 (May 1972), pp. 337–342.
DOI: 10.1145/355602.361310.

[Gen72c] W. M. Gentleman. "Implementing Clenshaw-Curtis quadrature, II computing the cosine transformation". In: *Communications of the ACM* 15.5 (May 1972), pp. 343–346.
DOI: 10.1145/355602.361311.

[GMPS16] K. Goździewski, C. Migaszewski, F. Panichi, and E. Szuszkiewicz. "The Laplace resonance in the Kepler-60 planetary system". In: *Monthly Notices of the Royal Astronomical Society: Letters* 455.1 (1st Jan. 2016), pp. L104–L108.
DOI: 10.1093/mnrasl/slv156.

[GT71] R. Goscinny and J. Tabary. *Une carotte pour Iznogoud*. Dargaud, 1971.

[GV13] G. H. Golub and C. F. Van Loan. *Matrix Computations (4th Edition)*. Johns Hopkins University Press, 15th Feb. 2013.

[Hal09] E. Halley. "A new, exact, and easy Method of finding the Roots of any Equations generally, and that without any previous Reduction". In: *The Philosophical Transactions of the Royal Society of London, from their commencement, in 1665, to the year 1800; Abridged, with notes and biographic illustrations*. Ed. by C. Hutton, G. Shaw, and R. Pearson. Vol. III from 1683 to 1694. Translated from the Latin [Hal94]. 1809, pp. 640–649.

[Hal94] E. Halley. "Methodus Nova Accurata & Facilis Inveniendi Radices Æquationum quarumcumque generaliter, sine prævia Reductione". In: *Philosophical Transactions of the Royal Society* 18.210 (May 1694), pp. 136–148.
DOI: 10.1098/rstl.1694.0029. English translation: [Hal09].

[HEFS72] T. E. Hull, W. H. Enright, B. M. Fellen, and A. E. Sedgwick. "Comparing Numerical Methods for Ordinary Differential Equations". In: *SIAM Journal on Numerical Analysis* 9.4 (1972), pp. 603–637.
DOI: 10.1137/0709052.

[HHZ12] 何志洲, 黄乘利, and 张冕. "火星重力场模型发展回顾及对萤火一号的展望—On the Mars Gravity Field Models and Prospects of "Yinhuo-1" Mission". In: 天文学进展—*Progress in Astronomy* 30.2 (May 2012), pp. 220–235.
CNKI: TWJZ201202007.

[Hig02] N. J. Higham. *Accuracy and Stability of Numerical Algorithms*. Society for Industrial and Applied Mathematics, 2002.

[HLB08] Y. Hida, X. S. Li, and D. H. Bailey. "Library for Double-Double and Quad-Double Arithmetic". Preprint at https://www.davidhbailey.com/dhbpapers/qd.pdf. 8th May 2008.

[HLW03] E. Hairer, C. Lubich, and G. Wanner. "Geometric numerical integration illustrated by the Störmer–Verlet method". In: *Acta Numerica* (May 2003), pp. 399–450.
DOI: 10.1017/S0962492902000144.

[HMR08] E. Hairer, R. I. McLachlan, and A. Razakarivony. "Achieving Brouwer's law with implicit Runge–Kutta methods". In: *BIT Numerical Mathematics* 48.2 (2008), pp. 231–243.
DOI: 10.1007/s10543-008-0170-3.

[HMS09] E. Hairer, R. I. McLachlan, and R. D. Skeel. "On energy conservation of the simplified Takahashi–Imada method". In: *Mathematical Modelling and Numerical Analysis* 43.4 (2009). ID: unige:5211, pp. 631–644.

[HO09] D. Huybrechs and S. Olver. "Highly oscillatory quadrature". In: *Highly Oscillatory Problems*. Ed. by B. Engquist, A. Fokas, and E. Hairer. Cambridge University Press, 2009, pp. 25–50.
DOI: 10.1017/CBO9781139107136.

[Hou68] A. S. Householder. "Bigradients and the Problem of Routh and Hurwitz". In: *SIAM Review* 10.1 (Jan. 1968), pp. 56–66.
DOI: 10.1137/1010003.

[Hou70] A. S. Householder. *The Numerical Treatment of a Single Nonlinear Equation*. International Series in Pure and Applied Mathematics. McGraw-Hill, 1970.

[HPS14] J. Hoffstein, J. Pipher, and J. H. Silverman. *An Introduction to Mathematical Cryptography*. Undergraduate Texts in Mathematics. Springer, 2014.

[HS05] E. Hairer and G. Söderlind. "Explicit, Time Reversible, Adaptive Step Size Control". In: *SIAM Journal on Scientific Computing* 26.6 (2005), pp. 1838–1851.
DOI: 10.1137/040606995.

[HW10] E. Hairer and G. Wanner. *Linear multistep method*. Scholarpedia. Department of Mathematics, University of Geneva, Switzerland, 2010.
DOI: 10.4249/scholarpedia.4591.
eprint: https://www.scholarpedia.org/article/Linear_multistep_method.

[INO06] A. Iserles, S. P. Nørsett, and S. Olver. "Highly Oscillatory Quadrature: The Story so Far". In: *Numerical Mathematics and Advanced Applications*. Ed. by A. B. de Castro, D. Gómez, P. Quintela, and P. Salgado. Springer Berlin Heidelberg, 2006, pp. 97–118. DOI: 10.1007/978-3-540-34288-5_6.

[JA12] B. A. Jones and R. L. Anderson. "A Survey of Symplectic and Collocation Integration Methods for Orbit Propagation". In: *22nd AAS/AIAA Spaceflight Mechanics Meeting* (Charleston, South Carolina, 29th Jan.–2nd Feb. 2012). Advances in the Astronautical Sciences 143. AAS 12-214. American Astronautical Society. Univelt, 2012. eprint: http://hdl.handle.net/2014/42453.

[JE33] E. Jahnke and F. Emde. *Funktionentafeln mit Formeln und Kurven—Tables of functions with formulæ and curves*. Teubner, 1933.

[JE38] E. Jahnke and F. Emde. *Funktionentafeln mit Formeln und Kurven—Tables of functions with formulæ and curves*. Teubner, 1938.

[JEL60] E. Jahnke, F. Emde, and F. Lösch. *Tafeln Höherer Funktionen—Tables of higher functions*. Teubner, 1960.

[JMP17] M. Joldes, J.-M. Muller, and V. Popescu. "Tight and Rigorous Error Bounds for Basic Building Blocks of Double-Word Arithmetic". In: *ACM Transactions on Mathematical Software* 44.2 (Oct. 2017), pp. 1–27. DOI: 10.1145/3121432.

[Jon12] B. Jones. "Orbit Propagation Using Gauss-Legendre Collocation". In: *AIAA/AAS Astrodynamics Specialist Conference* (Minneapolis, Minnesota, 13th–16th Aug. 2012). Vol. 2. Guidance, Navigation, and Control and Co-located Conferences. AIAA 2012-4967. American Institute of Aeronautics and Astronautics. Curran, 2012, pp. 1570–1585. DOI: 10.2514/6.2012-4967.

[Kah06] W. Kahan. "How Futile are Mindless Assessments of Roundoff in Floating-Point Computation ?" Rant at https://people.eecs.berkeley.edu/~wkahan/Mindless.pdf. 11th Jan. 2006.

[KB01] W. Kahan and D. Bindel. "Computing a Real Cube Root". 2001 retypesetting by Bindel of a purported 1991 version by Kahan, at https://csclub.uwaterloo.ca/~pbarfuss/qbrt.pdf. 21st Apr. 2001.

[Kep09] J. Kepler. *Astronomia nova αιτιολογητος seu physica coelestis tradita commentariis de motibus stellæ martis ex observationibus G. V. Tichonis Brahe*. Jussu & sumptibus Rudolphi II romanorum imperatoris &c, 1609. DOI: 10.3931/e-rara-558.

[KJL08] T. Kowalkowski, J. Johannesen, and T. Lam. "Launch Period Development for the Juno Mission to Jupiter". In: AIAA/AAS Astrodynamics Specialist Conference and Exhibit (Honolulu, Hawaii, 18th–21st Aug. 2008). Guidance, Navigation, and Control and Colocated Conferences. AIAA 2008-7369. American Institute of Aeronautics and Astronautics, 21st Aug. 2008. DOI: 10.2514/6.2008-7369.

[KM13] A. Krishnamoorthy and D. Menon. "Matrix Inversion Using Cholesky Decomposition". In: *Signal Processing: Algorithms, Architectures, Arrangements, and Applications* (Poznań, Poland, 26th–28th Sept. 2013). IEEE, 2013, pp. 70–72.

[Knu98] D. E. Knuth. *The Art of Computer Programming, Volume 2, Seminumerical Algorithms*. 3rd ed. Addison-Wesley, 1998.

[Koz62] Y. Kozai. "Secular Perturbations of Asteroids with High Inclination and Eccentricity". In: *The Astronomical Journal* 67.9 (Nov. 1962), pp. 591–598. DOI: 10.1086/108790.

[KS05] S. Kucherenko and Y. Sytsko. "Application of Deterministic Low-Discrepancy Sequences in Global Optimization". In: *Computational Optimization and Applications* 30 (Mar. 2005), pp. 297–318. DOI: 10.1007/s10589-005-4615-1.

[Kud07] S. M. Kudryavtsev. "Long-term harmonic development of lunar ephemeris". In: *Astronomy & Astrophysics* 471 (2007), pp. 1069–1075. DOI: 10.1051/0004-6361:20077568.

[Kut01] W. Kutta. “Beitrag zur näherungsweisen Integration totaler Differentialgleichungen”. In: *Zeitschrift für Mathematik und Physik* 46 (Nov. 1901), pp. 435–453.

[Lag67] J.-L. Lagrange. *Œuvres de Lagrange*. Ed. by J.-A. Serret and G. Darboux. 14 vols. Gauthier-Villars, 1867–1892.
eprint: https://gallica.bnf.fr/ark:/12148/bpt6k2155691.

[Lag71] J.-L. Lagrange. “Sur le problème de Kepler”. In: *Mémoires de l’Académie royale des sciences et belles-lettres de Berlin* XXV, année 1769 (1771), pp. 204–233.

[Lap72] P.-S. de Laplace. “Recherches sur le calcul intégral et sur le système du monde”. In: *Histoire de l’Académie royale des sciences avec les mémoires de mathématique et de physique tirés des registres de cette Académie* 1792.2 (1772), pp. 267–376.
eprint: https://gallica.bnf.fr/ark:/12148/bpt6k35711/f433.

[Lap78] P.-S. de Laplace. *Œuvres complètes de Laplace*. 14 vols. Gauthier-Villars, 1878–1912.
eprint: https://gallica.bnf.fr/ark:/12148/bpt6k775861.

[Lar11] M. Lara. “Design of long-lifetime lunar orbits: A hybrid approach”. In: *Acta Astronautica* 69 (29th Mar. 2011), pp. 186–199.
DOI: 10.1016/j.actaastro.2011.03.009.

[LC61] M. L. Lidov. and H. F. Cleaves. “The evolution of orbits of artificial satellites of planets under the action of gravitational perturbations of external bodies”. In: *Planetary and Space Science* 9.10 (Oct. 1961). Translated from the Russian [Лид61], pp. 719–759.
DOI: 10.1016/0032-0633(62)90129-0.

[LDV04] V. Lainey, L. Duriez, and A. Vienne. “New accurate ephemerides for the Galilean satellites of Jupiter”. In: *Astronomy & Astrophysics* 420.3 (2004), pp. 1171–1183.
DOI: 10.1051/0004-6361:20034565.

[Lea86] W. M. Lear. *The Gravitational Acceleration Equations*. Internal Note 86-FM-15. NASA Johnson Space Center, Apr. 1986.

[Lea87] W. M. Lear. *The Programs TRAJ1 and TRAJ2*. Internal Note 87-FM-4. NASA Johnson Space Center, Apr. 1987.

[Lin81] S. Linnainmaa. “Software for Doubled-Precision Floating-Point Computations”. In: *ACM Transactions on Mathematical Software* 7.3 (Sept. 1981), pp. 272–283.
DOI: 10.1145/355958.355960.

[LJVF89] A. C. Long, J. J. O. Cappellari, C. E. Velez, and A. J. Fuchs. *Goddard Trajectory Determination System (GTDS) Mathematical Theory Revision 1*. Tech. rep. FDD/552-89/001 CSC/TR-89/6001. Computer Sciences Corporation, National Aeronautics, and Space Administration/Goddard Space Flight Center, July 1989.

[LLL12] 李松明, 李岩, and 李劲东. “‘天绘一号’传输型摄影测量与遥感卫星—Mapping Satellite-1 transmission type photogrammetric and remote sensing satellite”. In: 遥感学报—*Journal of Remote Sensing* 16.S1 (2012), pp. 10–16.
DOI: 10.11834/jrs.20120003.
CNKI: YGXB2012S1005.

[LM00] T. Lang and J.-M. Muller. *Bound on Run of Zeros and Ones for Images of Floating-Point Numbers by Algebraic Functions*. RR-4045. Institut national de recherche en informatique et en automatique, 2000.
eprint: https://hal.inria.fr/inria-00072593.

[Lov22] H. P. Lovecraft. “Herbert West–Reanimator”. In: *Home Brew* 1.1–6 (Feb.–July 1922).

[MA92] R. I. McLachlan and P. Atela. “The Accuracy of Symplectic Integrators”. In: *Nonlinearity* 5 (1992), pp. 541–562.
DOI: 10.1088/0951-7715/5/2/011.

[Mac15] B. D. Macomber. “Enhancements to Chebyshev-Picard iteration efficiency for generally perturbed orbits and constrained dynamical systems”. PhD thesis. 2015.

[Mar00] P. Markstein. *IA-64 and Elementary Functions: Speed and Precision*. Prentice Hall, 2000.

[McL02] R. I. McLachlan. “Families of High-Order Composition Methods”. In: *Numerical Algorithms* 31.1-4 (2002), pp. 233–246.
DOI: 10.1023/A:1021195019574.

[McL07] R. I. McLachlan. “A New Implementation of Symplectic Runge–Kutta Methods”. In: *SIAM Journal on Scientific Computing* 29.4 (2007), pp. 1637–1649.
DOI: 10.1137/06065338X.

[McL93] R. I. McLachlan. “Symplectic integration of Hamiltonian wave equations”. In: *Numerische Mathematik* 66.1 (1993), pp. 465–492.
DOI: 10.1007/BF01385708.

[McL95] R. I. McLachlan. “On the Numerical Integration of Ordinary Differential Equations by Symmetric Composition Methods”. In: *SIAM Journal on Scientific Computing* 16.1 (Jan. 1995), pp. 151–168.
DOI: 10.1137/0916010.

[Mee98] J. Meeus. *Astronomical Algorithms*. 2nd ed. Willmann-Bell, 1998.

[Mon92] O. Montenbruck. “Numerical integration methods for orbital motion”. In: *Celestial Mechanics and Dynamical Astronomy* 53.1 (1992), pp. 59–69.
DOI: 10.1007/BF00049361.

[MQ02] R. I. McLachlan and G. R. W. Quispel. “Splitting methods”. In: *Acta Numerica* 11 (Jan. 2002), pp. 341–434.
DOI: 10.1017/S0962492902000053.

[MQ06] R. I. McLachlan and G. R. W. Quispel. “Geometric Integrators for ODEs”. In: *Journal of Physics A* 39 (2006), pp. 5251–5285.
DOI: 10.1088/0305-4470/39/19/S01.

[MR22] J.-M. Muller and L. Rideau. “Formalization of Double-Word Arithmetic, and Comments on “Tight and Rigorous Error Bounds for Basic Building Blocks of Double-Word Arithmetic””. In: *ACM Transactions on Mathematical Software* 48.1 (Feb. 2022), pp. 1–24.
DOI: 10.1145/3484514.

[Mul+10] J.-M. Muller, N. Brisebarre, F. De Dinechin, C.-P. Jeannerod, V. Lefèvre, G. Melquiond, N. Revol, D. Stehlé, and S. Torres. *Handbook of Floating-Point Arithmetic*. Birkhäuser, 2010.

[Mul16] J.-M. Muller. *Elementary Functions: Algorithms and Implementation, Third Edition*. Birkhäuser, 2016.

[Mul97] J.-M. Muller. *Elementary Functions: Algorithms and Implementation*. Birkhäuser, 1997.

[Mur98] A. Murua. “Runge–Kutta–Nyström methods for general second order ODEs with application to multi-body systems”. In: *Applied Numerical Mathematics* 28.2-4 (Oct. 1998), pp. 387–399.
DOI: 10.1016/S0168-9274(98)00055-5.

[Myr07] V. Myrnyy. “A Simple and Efficient FFT Implementation in C++”. In: *Dr. Dobb’s* (2007).
eprint: https://drdobbs.com/cpp/a-simple-and-efficient-fft-implementatio/199500857.

[New87] I. Newton. *Philosophiæ naturalis principia mathematica*. Facsimile at https://cudl.lib.cam.ac.uk/view/PR-ADV-B-00039-00001/1. Jussu Societatis Regiæ ac Typis Joseph Streater, 1687.

[New89] X. X. Newhall. “Numerical Representation of Planetary Ephemerides”. In: *Celestial Mechanics* 45 (1989), pp. 305–310.
eprint: http://adsabs.harvard.edu/full/1989CeMec..45..305N.

[Nie04] Y. Nievergelt. “Analysis and Applications of Priest’s Distillation”. In: *ACM Transactions on Mathematical Software* 30.4 (Dec. 2004), pp. 402–433.
DOI: 10.1145/1039813.1039815.

[NS09] P. Q. Nguyễn and D. Stehlé. “An LLL Algorithm with Quadratic Complexity”. In: *SIAM Journal on Computing* 39.3 (2009), pp. 874–903.
DOI: 10.1137/070705702.

[NW06] J. Nocedal and S. J. Wright. *Numerical Optimization*. Springer Series in Operations Research and Financial Engineering. Springer, 2006.

[NW86] A. M. Nobili and C. M. Will. “The real value of Mercury’s perihelion advance”. In: *Nature* 320 (6th Mar. 1986), pp. 39–41.
DOI: 10.1038/320039a0.

[OLBC10] F. Olver, D. Lozier, R. Boisvert, and C. Clark. *NIST Handbook of Mathematical Functions*. Cambridge University Press, 2010.

[OS94] D. I. Okunbor and R. D. Skeel. “Canonical Runge–Kutta–Nyström methods of orders five and six”. In: *Journal of Computational and Applied Mathematics* 51.3 (1994), pp. 375–382.
DOI: 10.1016/0377-0427(92)00119-T.

[Ove65] K. J. Overholt. “An Instability in the Fibonacci and Golden Section Search Methods”. In: *BIT* 5.4 (1965). Merged with the preceding article (starting p. 282) on the journal website, pp. 284–286.
DOI: 10.1007/BF01937508.

[Pao10] G. Paoloni. *How to Benchmark Code Execution Times on Intel® IA-32 and IA-64 Instruction Set Architectures*. White Paper 324264-001. Intel Corporation, Sept. 2010.

[PD81] P. J. Prince and J. R. Dormand. “High order embedded Runge-Kutta formulae”. In: *Journal of Computational and Applied Mathematics* 7.1 (1981), pp. 67–75.
DOI: 10.1016/0771-050X(81)90010-3.

[PL10] G. Petit and B. Luzum. *IERS Conventions (2010)*. IERS Technical Note 36. International Earth Rotation and Reference Systems Service Convention Centre, 2010.
eprint: http://www.iers.org/IERS/EN/Publications/TechnicalNotes/tn36.html.

[PR14] E. Pellegrini and R. P. Russell. “F and G Taylor Series Solutions to the Circular Restricted Three Body Problem”. In: *24th AAS/AIAA Spaceflight Mechanics Meeting* (Santa Fe, New Mexico, 26th–30th Jan. 2014). Advances in the Astronautical Sciences 152. AAS 14-237. American Astronautical Society. Univelt, 2014.

[QT90] G. D. Quinlan and S. Tremaine. “Symmetric multistep methods for the numerical integration of planetary orbits”. In: *Astronomical Journal* 100 (Nov. 1990), pp. 1694–1700.

[Qui99] G. D. Quinlan. “Resonances and instabilities in symmetric multistep methods”. Preprint at https://arxiv.org/pdf/astro-ph/9901136.pdf. 12th Jan. 1999.

[QZJ08] 曲宏松, 张叶, and 金光. “基于 Q 值选取的太阳同步回归轨道设计算法—Repeat sun-synchronous orbit design method based on Q value selection”. In: 光学精密工程—*Optics and Precision Engineering* 16.9 (Sept. 2008), pp. 1688–1694.
CNKI: GXJM200809026.

[Rap90] J. Raphson. *Analysis Æquationum Universalis seu Ad Æquationes Algebricas Resolvendas Methodus Generalis, et Expedita, Ex nova Infinitarum serierum Doctrina Deducta ac Demonstrata*. Prostant venales apud Abelem Swalle, ad Insigne Monocerotis in Cœmeterio Divi Pauli, 1690.
eprint: https://books.google.co.uk/books?id=A8NlAAAAcAAJ.

[Rie+16] J. Ries, S. Bettadpur, R. Eanes, Z. Kang, U. Ko, C. McCullough, P. Nagel, N. Pie, S. Poole, H. Save, and B. Tapley. *The Combination Global Gravity Model GGM05C*. Technical Memorandum CSR-TM-16-01. Center for Space Research at the University of Texas at Austin, Jan. 2016.
eprint: ftp://ftp.csr.utexas.edu/pub/grace/GGM05/README_GGM05C.pdf.

[RL06] R. P. Russell and M. Lara. “Repeat Ground Track Lunar Orbits in the Full-Potential Plus Third-Body Problem”. In: *A collection of technical papers*. AIAA/AAS Astrodynamics Specialist Conference and Exhibit (Keystone, Colorado, 21st–24th Aug. 2006). Vol. 3. Guidance, Navigation, and Control and Co-located Conferences. AIAA 2006-6750. American Institute of Aeronautics and Astronautics, 24th Aug. 2006, pp. 1935–1954.
DOI: 10.2514/6.2006-6750.

[RT87a] A. H. G. Rinnooy Kan and G. T. Timmer. “Stochastic global optimization methods – part I: Clustering methods”. In: *Mathematical Programming* 39 (Sept. 1987), pp. 27–56.
DOI: 10.1007/BF02592070.

[RT87b] A. H. G. Rinnooy Kan and G. T. Timmer. “Stochastic global optimization methods – part II: Multi level methods”. In: *Mathematical Programming* 39 (Sept. 1987), pp. 57–78.
DOI: 10.1007/BF02592071.

[Rut83] R. Ruth. "A Canonical Integration Technique". In: *Proceedings of the 1983 Particle Accelerator Conference (PAC 83)* (Santa Fe, New Mexico, 21st–23rd Mar. 1983). IEEE Transactions on Nuclear Science 30.4. Aug. 1983, pp. 2669–2671.
DOI: 10.1109/TNS.1983.4332919.
eprint: https://accelconf.web.cern.ch/accelconf/p83/PDF/PAC1983_2669.PDF.

[Sat69] Saturn V Flight Evaluation Working Group. *Saturn V Launch Vehicle, Flight Evaluation Report AS–503, Apollo 8 Mission.* Tech. rep. MPR-SAT-FE-69-1. George C. Marshall Space Flight Center, Feb. 1969.
eprint: http://hdl.handle.net/2060/19690015314.

[Sch70] E. Schröder. "Ueber unendlich viele Algorithmen zur Auflösung der Gleichungen". In: *Mathematische Annalen* 2 (June 1870), pp. 317–365.
DOI: 10.1007/BF01444024. English translation: [SS93].

[Sch73] L. Schrön. *Tables de logarithmes à sept décimales, pour les nombres depuis 1 jusqu'à 108000 et pour les fonctions trigonométriques de dix en dix secondes.* Gauthier-Villars, 1873.
eprint: https://books.google.fr/books?id=N2a87ajKxIIC.

[SG01] P. Sebah and X. Gourdon. "Newton's method and high order iterations". In: *Numbers, constants and computation.* 3rd Oct. 2001.
eprint: http://numbers.computation.free.fr/Constants/Algorithms/newton.html.

[SM10] J. R. Scott and M. C. Martini. "High-Speed Solution of Spacecraft Trajectory Problems Using Taylor Series Integration". In: *Journal of Spacecraft and Rockets* 47.1 (2010), pp. 199–202.

[Smi61] O. K. Smith. "Eigenvalues of a Symmetric 3 × 3 Matrix". In: *Communications of the ACM* 4.4 (Apr. 1961), p. 168.
DOI: 10.1145/355578.366316.

[SN97] M. Sidlichovsky and D. Nesvorny. "Frequency modified Fourier transform and its application to asteroids". In: *Celestial Mechanics and Dynamical Astronomy* 65 (1997), pp. 137–148.

[Som93] B. P. Sommeijer. "Explicit, high-order Runge–Kutta–Nyström methods for parallel computers". In: *Applied Numerical Mathematics* 13.1–3 (1993), pp. 221–240.
DOI: 10.1016/0168-9274(93)90145-H.

[SQG13] P. W. Sharp, M. A. Qureshi, and K. R. Grazier. "High order explicit Runge–Kutta Nyström pairs". In: *Numerical Algorithms* 62.1 (2013), pp. 133–148.
DOI: 10.1007/s11075-012-9571-0.

[SS02] M. Sofroniou and G. Spaletta. "Symplectic Methods for Separable Hamiltonian Systems". In: *Computational Science — ICCS 2002* (Amsterdam, The Netherlands, 21st–24th Apr. 2002). Ed. by P. M. A. Sloot, A. G. Hoekstra, C. J. K. Tan, and J. J. Dongarra. Lecture Notes in Computer Science 3. International Conference on Computational Science. Springer, Berlin, Heidelberg, 10th Apr. 2002, pp. 506–515.
DOI: 10.1007/3-540-47789-6_53.

[SS93] E. Schröder and G. W. Stewart. *On Infinitely Many Algorithms for Solving Equations.* Tech. rep. UMIACS-TR-92-121. Translated from the German [Sch70]. Institute for Advanced Computer Studies, University of Maryland, College Park, Jan. 1993.
eprint: http://hdl.handle.net/1903/577.

[ST95] T. R. Scavo and J. B. Thoo. "On the Geometry of Halley's Method". In: *The American Mathematical Monthly* 102.5 (May 1995), pp. 417–426.
DOI: 10.2307/2975033.

[Sta98] E. M. Standish. *JPL Planetary and Lunar Ephemerides, DE405/LE405.* Interoffice Memorandum IOM 312.F–98–048. Jet Propulsion Laboratory, Aug. 1998.

[Ste74] P. H. Sterbenz. *Floating-point computation.* Prentice-Hall, 1974.

[Stö12] C. Störmer. "Sur les trajectoires des corpuscules électrisés dans l'espace sous l'action du magnétisme terrestre, avec application aux aurores boréales". In: *Radium* 9.11 (Nov. 1912), pp. 395–399.
DOI: 10.1051/radium:01912009011039501.

[Suz90] M. Suzuki. "Fractal decomposition of exponential operators with applications to many-body theories and Monte Carlo simulations". In: *Physics Letters A* 146 (June 1990), pp. 319–323.
DOI: 10.1016/0375-9601(90)90962-N.

[SZ04] D. Stehlé and P. Zimmermann. *Gal's Accurate Tables Method Revisited*. Research Report RR-5359. INRIA, Oct. 2004.
eprint: https://inria.hal.science/inria-00070644v1/file/rr-5359.pdf.

[SZ05] D. Stehlé and P. Zimmermann. "Gal's accurate tables method revisited". In: *17th IEEE Symposium on Computer Arithmetic (ARITH'05)* (Cape Cod, MA, USA, 27th–29th June 2005). Ed. by P. Montuschi and E. Schwarz. IEEE Computer Society, June 2005, pp. 257–264.
DOI: 10.1109/ARITH.2005.24.

[Sze67] V. G. Szebehely. *Theory of Orbits. The Restricted Problem of Three Bodies*. Academic Press, 1967.

[SZG22] A. Sibidanov, P. Zimmermann, and S. Glondu. "The CORE-MATH Project". In: *2022 IEEE 29th Symposium on Computer Arithmetic (ARITH)*. IEEE, Sept. 2022, pp. 26–34.
DOI: 10.1109/ARITH54963.2022.00014.
eprint: https://inria.hal.science/hal-03721525v3/file/core-math-final.pdf.

[Tao12] T. Tao. *A mathematical formalisation of dimensional analysis*. Blog post. University of California, Los Angeles, 29th Dec. 2012.
eprint: https://terrytao.wordpress.com/2012/12/29/a-mathematical-formalisation-of-dimensional-analysis/.

[TBS13] Y. Takahashi, M. W. Busch, and D. J. Scheeres. "Spin state and moment of inertia characterization of 4179 Toutatis". In: *The Astronomical Journal* 146.4 (Sept. 2013), pp. 95–104.
DOI: 10.1088/0004-6256/146/4/95.

[Tre10] L. N. Trefethen. "Householder triangularization of a quasimatrix". In: *IMA Journal of Numerical Analysis* 30.4 (2010), pp. 887–897.
DOI: 10.1093/imanum/drp018.

[Tre97] L. N. Trefethen. *Maxims about numerical mathematics, science, computers, and life on Earth*. Maxims. Cornell University, 1997.
eprint: https://people.maths.ox.ac.uk/trefethen/maxims.html.

[Ver67] L. Verlet. "Computer "Experiments" on Classical Fluids. I. Thermodynamical Properties of Lennard-Jones Molecules". In: *Physical Review* 159.1 (5th July 1967), pp. 98–103.
DOI: 10.1103/PhysRev.159.98.

[Wal18] U. Walter. *Astronautics – The Physics of Space Flight – Third Edition*. Springer Nature Switzerland AG, 2018.
DOI: 10.1007/978-3-319-74373-8.

[Wal85] J. Wallis. *A Treatise of Algebra, both Historical and Practical. Shewing, the Original, Progress, and Advancement thereof, from time to time; and by what Steps it hath attained to the Heighth at which now it is*. Richard Davis, 1685.
eprint: https://books.google.fr/books?id=TXpmAAAAcAAJ.

[Wan43] L. Wantzel. "Classification des nombres incommensurables d'origine algébrique". In: *Nouvelles annales de mathématiques. Journal des candidats aux écoles polytechnique et normale* 1.2 (1843). Ed. by O. Terquem and C.-C. Gerono, pp. 117–127.
eprint: http://www.numdam.org/item/?id=NAM_1843_1_2__117_1.

[War03] H. S. Warren Jr. *Hacker's Delight*. Addison-Wesley, 2003.

[Wes17] D. B. Westra. *Identites and properties for associated Legendre functions*. Personal note. Universität Wien, Sept. 2017.
eprint: https://www.mat.univie.ac.at/~westra/associatedlegendrefunctions.pdf.

[WG13] G. Wilkins and M. Gu. "A modified Brent's method for finding zeros of functions". In: *Numerische Mathematik* 123 (2013), pp. 177–188.
DOI: 10.1007/s00211-012-0480-x.

[WL17] 汪海洪 and 罗北. “计算测高卫星地面轨迹交叉点的快速数值算法—Fast Numerical Algorithm for the Calculation of Altimetric Crossovers from Satellite Ground Tracks”. In: 武汉大学学报 · 信息科学版—*Geomatics and Information Science of Wuhan University* 42.3 (Mar. 2017), pp. 293–298.
DOI: 10.13203/j.whugis20140866.
CNKI: WHCH201703002.

[WYZ15] 温生林, 闫野, and 张华. “低轨回归轨道卫星轨迹漂移特性分析与控制—Analysis and control of groundtrack drift for recursive low earth orbit satellites”. In: 系统工程与电子技术—*Systems Engineering and Electronics* 37.3 (2015). The DOI is not registered; see https://www.sys-ele.com/CN/10.3969/j.issn.1001-506X.2015.03.22, pp. 613–619.
DOI: 10.3969/j.issn.1001-506X.2015.03.22.
CNKI: XTYD201503023.

[WZ91] X. Wu and K. Zhang. “A better tree-structured vector quantizer”. In: *1991 Data Compression Conference* (Snowbird, UT, USA, 8th–11th Apr. 1991). Ed. by J. A. Storer and J. H. Reif. IEEE Computer Society, Apr. 1991, pp. 392–401.
DOI: 10.1109/DCC.1991.213341.

[XZL09] 徐莹, 张有广, and 林明森. “卫星高度计轨道设计的因素分析—An Analysis on the Considerations of Satellite Altimeter Orbit Design”. In: 遥感技术与应用—*Remote Sensing Technology and Application* 24.2 (2009), pp. 155–163.
DOI: 10.11873/j.issn.1004-0323.2009.2.155.
CNKI: YGJS200902005.

[Yos90] H. Yoshida. “Construction of higher order symplectic integrators”. In: *Physics Letters A* 150.5–7 (1990), pp. 262–268.
DOI: 10.1016/0375-9601(90)90092-3.

[Zha+15] Y. Zhao, J. Ji, J. Huang, S. Hu, X. Hou, Y. Li, and W.-H. Ip. “Orientation and rotational parameters of asteroid 4179 Toutatis: new insights from Chang’e-2’s close flyby”. In: *Monthly Notices of the Royal Astronomical Society* 450.4 (May 2015), pp. 3620–3632.
DOI: 10.1093/mnras/stv792.

[Zhō+12] 周飞, 李强, 信太林, 韦锡峰, and 张华. “空间辐射环境引起在轨卫星故障分析与加固对策—Analyses and countermeasures of in-orbit satellite failures caused by space radiation environment”. In: 航天器环境工程—*Spacecraft Environment Engineering* 29.4 (2012), pp. 392–396.
DOI: 10.3969/j.issn.1673-1379.2012.04.006.
CNKI: HTHJ201204009.

[ZJL02] K. Zatloukal, M. H. Johnson, and R. E. Ladner. “Nearest Neighbor Search for Data Compression”. In: *Near Neighbor Searches, and Methodology: Fifth and Sixth DIMACS Implementation Challenges*. Ed. by M. H. Goldwasser, D. S. Johnson, and C. C. McGeoch. American Mathematical Society, 2002, pp. 69–86.

[ZSG+24] P. Zimmermann, A. Sibidanov, S. Glondu, et al. *The CORE-MATH Project*. Software. Apr. 2024.

[Афа06] И. Афанасьев. “Реабилитация «Рокота» открывает дорогу новым миссиям ЕКА”. In: *Новости космонавтики* 16.1 (Jan. 2006), p. 7.

[Дуб+01] Я. В. Дубровинский, К. В. Журавлев, А. С. Исаихин, О. Н. Молоканова, and Н. И. Шершнева. “Итоги баллистического обеспечения космических аппаратов «Метеор-Природа», «Ресурс-О1» и «Электро»”. In: *Вопросы электромеханики. Труды ВНИИЭМ* 100 (2001), pp. 54–69.

[Лид61] М. Л. Лидов. “Эволюция орбит искусственных спутников планет под действием гравитационных возмущений внешних тел”. In: *Искусственные Спутники Земли* 8 (1961), pp. 5–45. English translation: [LC61].